

ECE 4375 Homework Assignments

1. Given:

$$2\ddot{y}(t) + 7\dot{y}(t) + 3y(t) - 4\dot{u}(t) - 3u(t) = 0,$$

assuming all the initial conditions are zero.

Find:

- A. The transfer function, $\frac{Y(s)}{U(s)}$
- B. The zeros and poles

2. Given:

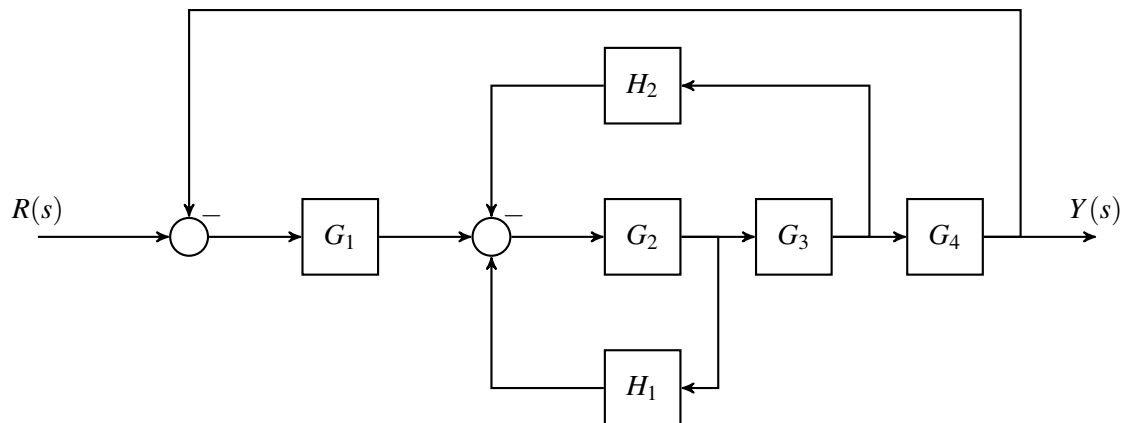
$$\ddot{y}(t) - 8\dot{y}(t) + 2\ddot{u}(t) - 7\dot{u}(t) = 3u(t),$$

assuming all the initial conditions are zero.

Find:

- A. The transfer function, $\frac{Y(s)}{U(s)}$
- B. The zeros and poles

3. Given:

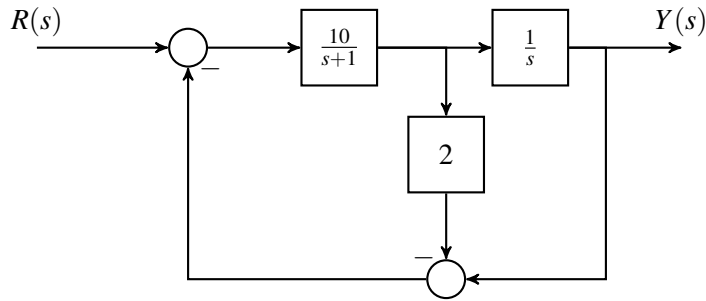


Find:

- A. open-loop transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$
- B. use the solution in A to find closed-loop transfer functions G and H where,

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)}$$

4. Given:



Find:

- A. open-loop transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$
- B. closed-loop transfer functions G and H where, $\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)}$

5. Given:

$$\frac{Y(s)}{U(s)} = \frac{4(s+50)}{s^2+30s+200}$$

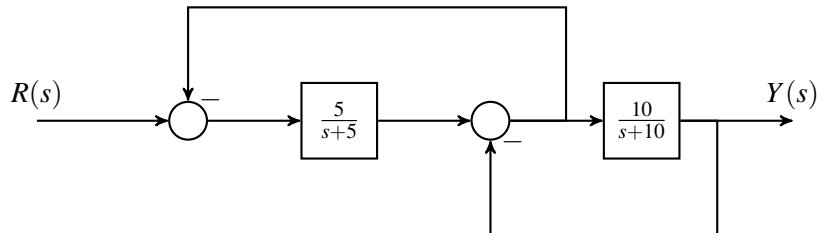
- A. Find $y(\infty)$ when the input is:
 1. $u(t)$ is an impulse
 2. $u(t)$ is a unit step
- B. Determine the block diagram of the equation

6. Given:

$$\frac{Y(s)}{U(s)} = \frac{s^2-4s+6}{s^2+3s+6}$$

- A. Find $y(\infty)$ when the input is:
 1. $u(t)$ is an impulse
 2. $u(t)$ is a unit step
- B. Find the block diagram of the equation

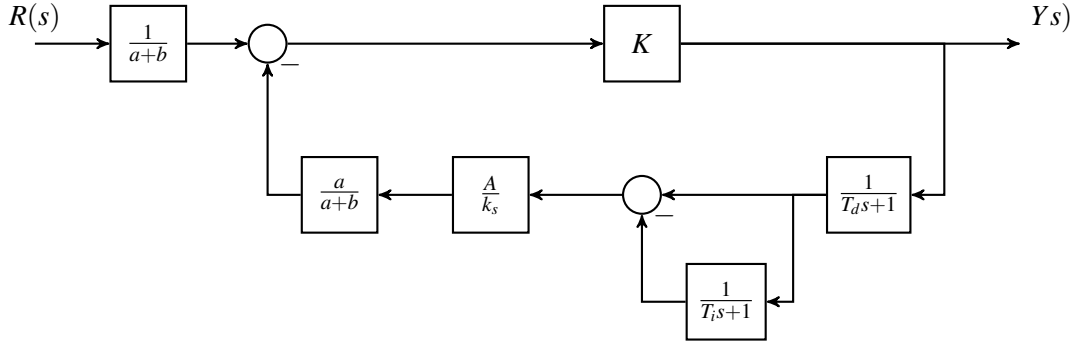
7. Given:



Find:

- A. The transfer function $\frac{Y(s)}{R(s)}$
- B. Determine stability
- C. closed-loop transfer functions G and H where, $\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)}$

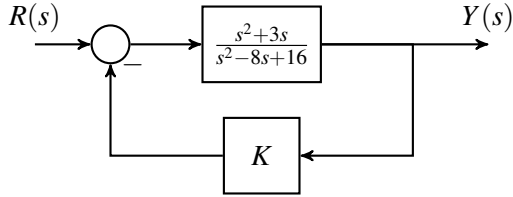
8. Given:



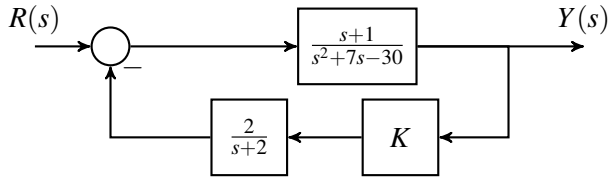
Find:

- A. The transfer function, $\frac{Y(s)}{R(s)}$
- B. Set: $a = 1$, $b = 3$, $T_i = 10$, $T_d = 1$, $A = 2$, $k_s = 4$, and $K = 1$ then find the poles and zeros of the system.

9. Given the system, where $K > 0$, find the range of K such that the system is stable using the Routh Array



10. Given the system, where $K > 0$, find the range of K such that the system is stable using the Routh Array



11. Sketch the Root Locus for the following loop-gain equations (assuming a typical negative feedback block diagram),

- A. $GH(s) = \frac{K}{s(s+10)(s+8)}$
- B. $GH(s) = \frac{K}{(s+2)(s^2+2s+2)}$
- C. $GH(s) = \frac{K(s+5)}{s(s+1)(s+10)}$
- D. $GH(s) = \frac{K(s^2+4s+8)}{s^2(s+1)}$

Where appropriate find:

1. σ_i

2. Θ_k
3. break away, BA, and break in, BI points
4. gain K at the BA and BI points
5. critical gain for stability, K_{cr}

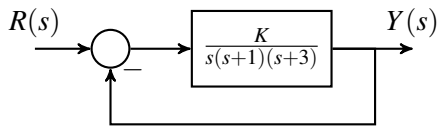
12. Given the system $G(s)$ with unity-negative feedback

$$G(s) = \frac{4}{s(s+5)}$$

Find

- A. $y(\infty)$ for a set-point input
- B. $y(0)$ for a set-point input
- C. $y(t)$ for a impulse input

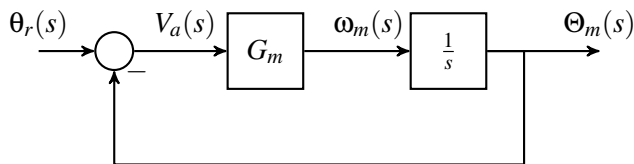
13. Determine the range of K such that the closed-loop system is stable, when



14. Given a position control system using a DC-Servo motor with the following parameters

$$\bullet K_m = 2.5, L_a = 0, K_b = 0.1, J_m = 0.4, J_L = 9.6, B = 0.23, R_a = 2.3$$

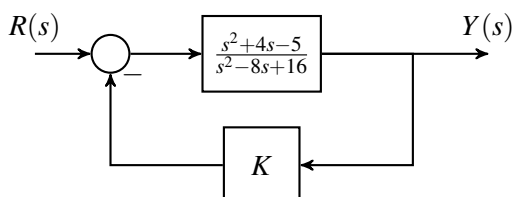
where the describing diagram is,



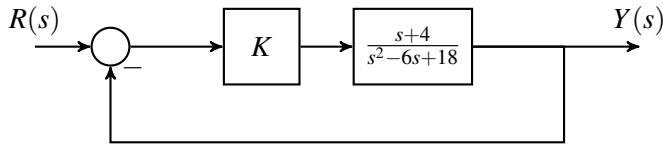
Find

- A. The transfer function of the motor only, $\frac{\omega_m(s)}{V_a(s)}$
- B. The transfer function of the motor and integrator only, $\frac{\theta_m(s)}{V_a(s)}$
- C. The transfer function of the complete system with feedback, $\frac{\theta_m(s)}{\theta_r(s)}$
- D. Determine the stability for each transfer function

15. Given the system, where $K > 0$, find the range of K such that the system is stable using the Routh Array



16. Given the system, where $K > 0$, find the range of K such that the system is stable using the Routh Array



17. Consider the following characteristic equation:

$$A(s) = s^4 + s^3 + K_1 s^2 + s + K_2 = 0$$

Determine the range of stability for K_1 and K_2 .

18. Given a typical negative feedback model and the following transfer functions:

A. $G(s) = \frac{s+1}{s^2(s+10)}, H(s) = 1$

B. $G(s) = \frac{10}{s(s+2)}, H(s) = \frac{1}{s+1}$

C. $G(s) = \frac{s+10}{(s^2+0.1s+1)^2}, H(s) = 1$

D. $G(s) = \frac{(s+1)^2}{s(s+10)}, H(s) = 1$

E. $G(s) = \frac{5s^4+80s^3+510s^2+1620s+2160}{s^4+2.5s^3+1.5s^2}, H(s) = \frac{5}{s+5}$

Find: (1) BIBO stability, (2) frequency response analysis (Bode Plot with gain and phase margin identified), (3) the steady-state error for position, velocity, and acceleration

19. Sketch the root locus for the following characteristic equations.

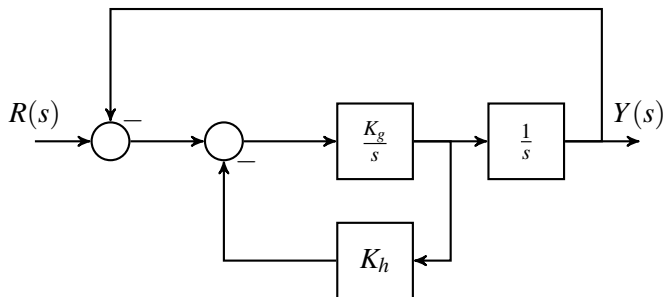
A. $1 + \frac{K}{s(s+10)(s+8)} = 0$

B. $1 + \frac{K}{(s+2)(s^2+2s+2)} = 0$

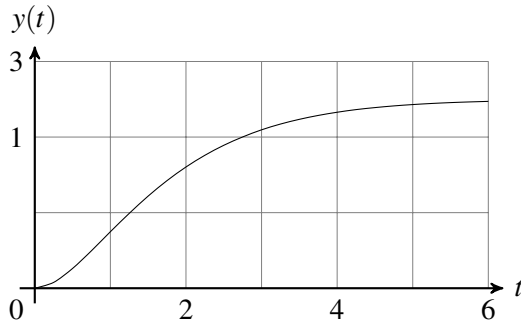
C. $1 + \frac{K(s+5)}{s(s+1)(s+10)} = 0$

D. $1 + \frac{K(s^2+4s+8)}{s^2(s+1)} = 0$

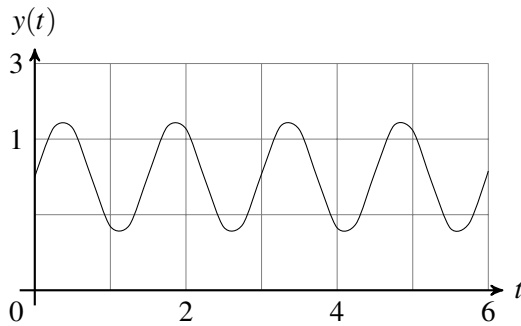
20. Given the block diagram, find the values of K_g and K_h such that the maximum overshoot to a unit step input is 25% and the peak time t_p is specified as 2 seconds.



21. Given the open-loop output response, label the figure and determine the Ziegler-Nichols 1st method PID Table.



22. Given the closed-loop output response, label the figure and determine the Ziegler-Nichols 2nd method PID Table.



23. Draw a Bode diagram for the following transfer functions:

A. $G(s) = \frac{10(s+3)}{s(s+400)(s+2000)}$

B. $G(s) = \frac{s(s+10)(s+2000)}{(s^2+1001s+1000)(s+100)}$

C. $G(s) = \frac{20(s+100)}{2(s+1)(s+10)}$

24. Use MATLAB to generate a Bode Diagram and determine the gain and phase margins for the following transfer functions:

A. $G(s) = \frac{300(s+200)}{s^2(s+50)(s+20)}$

B. $G(s) = \frac{2.5 \times 10^9(s+10)}{(s^2+2500s+2.5 \times 10^7)(s+100)}$

C. $G(s) = \frac{(s^2+5s+10000)}{10000(s+1)(s+10)}$

25. Given a DC-Servo motor, with tachometer feedback K_f , feedforward control K_p , and the following parameters

- $K_m = 5, L_a = 0.00002, K_b = 0.15, J_T = 1.25, B = 0.35, R_a = 3.3.$

Using software, simulate a position control setup such that in the input is an angular reference position and the output is the actual position.

Find

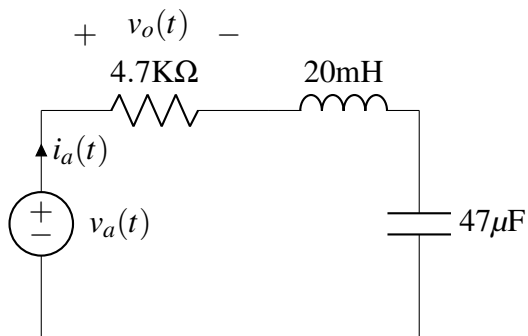
A. plot the step response when

1. $K_p = 1, K_f = 0$
2. $K_p = 1, K_f = 1$
3. $K_p = 5, K_f = 1$
4. $K_p = 25, K_f = 2.5$

B. Set $K_p = 1$ and $K_f = 1$ and determine

1. The open-loop transfer function, $\frac{\Theta(s)}{\Theta_{\text{ref}}(s)} = \frac{B(s)}{A(s)}$
2. The closed-loop transfer function, $\frac{\Theta(s)}{\Theta_{\text{ref}}(s)} = \frac{G(s)}{1+GH(s)}$

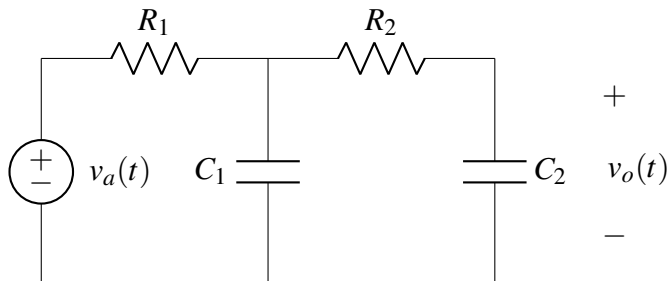
26. Given the circuit diagram



Find

- A. The transfer function, $\frac{V_o(s)}{V_a(s)}$
- B. The zeros and poles
- C. From A determine the loop-gain, $L(s)$
- D. The loop-gain zeros and poles

27. Given the circuit diagram



Find

- A. The transfer function, $\frac{V_o(s)}{V_a(s)}$
- B. The zeros and poles

- C. From A determine the loop-gain, $L(s)$
- D. The loop-gain zeros and poles

A Additional work

Additional practice from the textbook:

1. Problems 1.1 thru 1.9 (Assignment 1)
2. Problems 3.40 thru 3.44 (Assignment 2)
3. Problems 5.8, 5.9, 5.10, and 5.33 (Assignment 3)